

# Predicting Students' Employability using Machine Learning Approach

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**Abstract**—This study aims to apply an approach using machine learning for predicting students' employability. The researchers conducted a case study that involved 27,000 information (3000 observations and 9 features) of students' Mock Job Interview Evaluation Results, On-the Job Training (OJT) Student Performance Rating and General Point Average (GPA) of students enrolled in OJT course of School Year 2015 to School Year 2018. Three learning algorithms were used such as Decision Trees (DT), Random Forest (RF), and Support vector machine (SVM) in order to understand how students get employed. The three algorithms were evaluated through the performance matrix as accuracy measures, precision and recall measures, f1-score and support measures. During the experiments Support Vector machine (SVM) obtained 91.22% in accuracy measures which was significantly better than all of the learning algorithms, DT 85%, RF 84%. The learning curve produced during the experiment displays the training error results which were above the one for validation error while the validation curve displays the testing output where gamma was best at 10 to 100 in gamma 5. This concludes that the model produced with SVM was not underfit and over-fit. This study is very promising that lead to the researchers to be motivated to enhanced the process and to validate the produced predictive model for further study.

**Keywords**—Classification, Machine Learning, SVM, Random Forest, Decision Trees, Employment prediction

## I. INTRODUCTION

The Higher Education (HE) imparts an important role in the development of every nation's economy [1]. This strengthens the country by contributing to a reliable and qualified workforce to society. Higher Education is the foundation for several benefits such as: encouraging and nurturing the talent; it is also increases the quality of national human capital; and the core means of a nation's competitive status to upgrade and this lead to educational institutions to find better ways in improving the employability of their students [2]. In most institutions, the major issues of every student is the employability, foreseeing students' employability prior to job application can increase institutional placement proportion. To identify the weaknesses prior to interview to any company can support students to improve the areas where they were identified as need for improvements [3]. The education is becoming more and more employment-oriented. This become the trends of employment of graduating students

from any institution and become a key factor in the reputation of every institution in the Philippines. Foreseeing the employability of each student before graduation can aid students on the areas that they need to work extensively to the areas need for improvements.

Most of the researches and studies that have been published used the techniques on data mining to predict employability. Some of the techniques were ,Decision Tree, NaïveBayes and Support Vector Machine [4]. The Logistic Regression, K-Nearest Neighbor, Random forest, SVM (LinearSVC), Quadratic Discriminant Analysis (QDA), and Multi-class Ada Boosted were also used in data mining techniques in predicting employability [5].

This paper aims to developed a machine learning approach in predicting student's employability and analyze its skillset signals. To achieved the said objective , this paper will present the initial stage consist pre-processing technique before applying learning algorithms. This paper is in developing stage of a machine learning based model to predict student's employability. The researchers were motivated to conduct the study in the context of emergent areas such as institutional Intelligence or academic analytics for the enhancement and promotion of those identified skill sets that will contribute to the better employment of engineering students of the Technological Institute of the Philippines.

The sections were presented in the following sections: Section 2 were the Background, Section 3 presented the Proposed Method, Section 4 discussed the Case Study of Students' Employability ,Section 5 were the Results & Analysis, Section 6 were the Conclusion and Section 7 were Acknowledgement.

## II. BACKGROUND

This study was developed for the benefit of the career center of Technological Institute of the Philippines (TIP). TIP was one of the leading engineering school in the Philippines. The career center of TIP was the model career center in the Philippines that accommodates students' employability development and career linkages. Career center conducts Students Development Program (SDP) and Mock Job Interview Evaluation Results for those engineering students who intends to enroll in the On-the

Job Training course (OJT) for engineering programs. Mock Job Interview is one of the requirements of engineering students prior to deployment in the company for their internship. It is evident that internship datasets increases year after year and resulted to the growing population of educational datasets [6].

As Higher Education Institutions (HEIs) are increasingly held accountable for students' career outcomes and as competition for jobs in the labor market increases, institutions need to determine students who are most likely to be employed and who needs improvement. To determine the areas student need to improve before the student undergo the interview of any company [5].

Different studies conducted in predicting employability of students. Most of the studies were done using data mining techniques. The most widely used techniques in data mining are classification and prediction techniques. The study of Sapaat [7] designed predictive model using data mining techniques on Graduate Employability where datasets were gathered from the tracer study of the Ministry of Higher Education of Malaysia consisting of all the graduates of public and private institutions, polytechnics where the algorithms such as Decision Tree, Naive Bayes, Neural Network, Logistic Regression, Random Forest were commonly used in employment classification and prediction. WEKA were also used as a data mining tool to build predictive models. The methods such as CRISPDM (Cross Industry Standard Process for Data Mining) and KDD (Knowledge Discovery in Databases) were the prediction analysis techniques that have been done. In this study the decision tree were concluded as most appropriate algorithm to predict the employability of students. The highest accuracy was obtained by the classification J48 (variant of decision trees)[8]. Tajul[12] used five data mining algorithms; Naive Bayes, Logistic Regression, Multilayer Perceptron, KNearest Neighbor and J48 Decision Tree to determine whether the graduates can be employed, unemployed or in some instance cannot be determined six months after graduation, the proponent used the bayesian and decision Tree algorithm [12]. The Novel Neural Network (NN) were also used to predict the unemployment rate based on the information gathered from the web. Another study of Sapaat uses the datasets collected from the Alumni unit (AU) Examinations, Unit(EU), and curriculum unit (CU) resulted to logistic regression as the best classifier to predict whether the graduates will be employed in a private or public sector, unemployed or continue their education. Most of the studies conducted used Data mining techniques to predict employability[9]

In Mishra's[1] study, used the built predictive model for employment status of graduates of Khon Kaen University. In the said study the proponents used Averaged One Dependence Estimators (AODE) Averaged One Dependence Estimators with subsumption resolution (AODEsr), Bayesian methods were also used such as Naïve Bayesian Simple, Naïve Bayesian, and Bayesian networks, and Naïve Bayesian Updateable. The highest accuracy results were evident in Averaged One-Dependence Estimators with subsumption resolution (AODEsr) algorithm which obtained 98.3% followed by the AODE which gathered 96.1% accuracy [1].

In the study conducted by Gao [10], he used the employment information as attributes. The classic decision tree classifiers were algorithms used for evaluation by doing the analysis and comparisons based on different criteria. Weka were used also by to build a data mining techniques to analyze his works. The study concludes that in for each gender the origin of students' place and every work is different, it shows that 50% of the graduates do not select the educational institution [10].

### III. PROPOSED METHOD

In this paper, the researchers followed a new approach in predicting employability of engineering students. In this method, the predictive models opened a new way to manage huge amounts of datasets.

Generally, the proposed method to predict students' employability (based on machine learning) uses the structure of data, orderly data approaches principles that were also used in data science [11]. Since, it is on its preliminary stage the proponent presented this study in the most comprehensible approach to explain its appropriateness [12].

The researcher proposed a new way of predicting students' employability using machine learning approach. Figure 1 explains the workflow of this proposal:

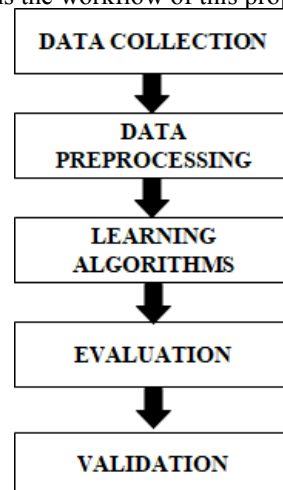


Fig. 1. Workflow of Experimental Set-Up

Figure 1 presented the workflow of this study, the first was collection of dataset of On the Job Training (OJT) students enrolled during 1st Semester of SY2015-2016 up to Summer SY2018-2019 OJT students information where there were 3,000 observations and nine (9) features and a total of 27,000 of information collected where 18,000 information of students or data collected from Mock Job Interview Results conducted by the career center and 3000 information of student performance ratings collected from the student performance rating or Student assessment tool rated by the supervisor of the students that serve as assessment tool used during their on-the job training (OJT) in the company and another 3000 observation from the General Point Grade of the students collected from Registrar's office, then the Data cleaning were performed and removed all observations who have only partial information. After that, the researchers normalized the data

before applying the learning algorithms. Then, merged the datasets collected from different offices. The merged datasets were used in the case study and those datasets collected from the conducted survey to the students who already graduated in SY 2015-2018 that were used to create the class, the learning algorithms used were related to supervised learning. Decision Trees (DT), Random Forest (RF) and Support Vector Machine (SVM) were the algorithms used by the researchers to predict students employability. The performance matrix used by the researchers to assess the performance of the models' accuracy, precision, recall, f1-score measures. The researchers compared the results of each learning algorithm and chose the model that has the highest accuracy among the three, the best predictive model that was built and can predict if the student is employable or less employable was Support Vector Machine (SVM).

#### IV. CASE STUDY: STUDENTS' EMPLOYABILITY

The data used in this study were gathered from the available information of students who completed a degree in Engineering programs from SY 2015- 2016 to SY2018-2019. The researchers conducted survey to those students graduated from the said School year. Then the class were created using the result of the survey. This is consisting of 27,000 information of students with 3000 observations and 9 features of each student. The researchers gathered the information by using two input methods such as data sets given by the career center of technological Institute of the Philippines and the raw data from the survey conducted by the researchers to the graduates of engineering programs and currently landed a job. The researchers used Google forms with privacy consent statement in survey deployment. The following are the process done by the researcher:

##### A. Data Collection

The datasets were collected from the Career Center of Technological Institute of the Philippines- Manila consists of Mock job Interview Results consist of three thousands (3000) observations and nine (9) features, Student Performance Rating of the OJT students collected by the On-The Job Training (OJT) Faculty In-charge and General Point Average from the Registrar's Office. The datasets that were collected were compliant of Data Privacy Act of the Philippines.

TABLE 1  
Students' Employability Dataset

| Feature Number | Description                |
|----------------|----------------------------|
| 1              | General Appearance         |
| 2              | Manner of Speaking         |
| 3              | Physical Condition         |
| 4              | Mental Alertness           |
| 5              | Self Confidence            |
| 6              | Ability to Present Ideas   |
| 7              | Communication Skills       |
| 8              | Student Performance Rating |
| 9              | General Point Grade        |

##### B. Data Pre-Processing

The following are the techniques used in pre-processing of datasets:

1. **Merging of Datasets**- The datasets collected from career center, Registrar's office and OJT Faculty In-charge were merged to have one dataset that will be used in the experiments [3].
2. **Data Normalization** : Each attribute or columns were filled with the median values when there is a missing values on attributes or columns. For row or number of observations were filled with the mean of that number of observation or row when there is a missing values for row[3].

##### C. Learning Algorithms

After performing the Feature Selection, the proponent split the data into training data into seventy percent (70%) and testing data into thirty percent (30%) respectively[1]. The researchers started training the datasets applying the learning algorithms are as follows:

1. Decision Trees – An algorithm that includes chance event outcomes, utility, and resource costs. The tree-like graph model can be noticeable and resulted into decisions and possible occurrences. Conditional control statements were used in this type of algorithm [3].
2. Random Forest – An algorithm that consists of numerous decision trees at a given period of training where each decision tree resulted into an individual output. Resultant output were produced in this type of algorithm. [3].
3. Support Vector Machine- This algorithm used the concept of decision planes that described decision boundaries[3].

##### D. Performance Evaluation

Table 2: Predictive Model Decision Trees Result

|                 | Accuracy | Precision | Recall | F1-Score | Support |
|-----------------|----------|-----------|--------|----------|---------|
| Employable      | 0.85     | 0.85      | 0.89   | 0.87     | 519     |
| Less-Employable | 0.84     | 0.84      | 0.78   | 0.81     | 376     |
| Avg/Total       | 0.845    | 0.85      | 0.85   | 0.85     | 895     |

Table 3: Predictive Model Random Forest Result

|                 | Accuracy | Precision | Recall | F1-Score | Support |
|-----------------|----------|-----------|--------|----------|---------|
| Employable      | 0.842    | 0.85      | 0.88   | 0.87     | 519     |
| Less-Employable | 0.841    | 0.83      | 0.78   | 0.81     | 376     |
| Avg/Total       | 0.84     | 0.84      | 0.84   | 0.84     | 895     |

Table 4: Predictive Model Support Vector Machine Result

|                 | Accuracy | Precision | Recall | F1-Score | Support |
|-----------------|----------|-----------|--------|----------|---------|
| Employable      | 0.912    | 0.91      | 0.911  | 0.911    | 519     |
| Less-Employable | 0.911    | 0.90      | 0.910  | 0.910    | 376     |
| Avg/Total       | 0.912    | 0.91      | 0.911  | 0.911    | 895     |

The following performance measures were used for performance evaluation of the three learning algorithms:

1. Precision – The precision is certainly the classifier's ability of not classifying a positive when the sample is negative. The score gathered was at its best value at 1 and worst score at 0. The ratio of true positives from the false positives can be shown as [12].

$$P = tp / (tp + fp) [12]$$

Where:

tp = true positives  
fp=false positives

1. Recall- It is the ability of the classifier to locate all the positive samples. This score was at its best value at 1 and its worst score at 0. The ratio of the true positives from the false negatives can be shown mathematically as [12].

$$R = tp / (tp + fn) \quad [12]$$

Where:

tp = true positive  
tn = true negative

2. F1-score- It is where 1 is the F1 score best value and 0 is its worst score . The relative contribution of precision and recall to the F1 score is equal. The weighted average of the precision and recall can be illustrated mathematically as [12].

$$F1\text{-score} = (P+R)/2 \quad [12]$$

4. The support can be shown as in each class in each predicted label , the number of occurrences were made [12].

Cross-evaluation performance measures were used consist of the following :

- 1) **Accuracy:** The correct predictions must be higher than incorrect prediction, this can be stated mathematically as:

$$Accuracy = (TP + TN) / (TP + FN + TN + FP) \quad [3] \quad (1)$$

- 2) **Sensitivity or Recall:** The proportion from instances of those are actually positive from correct positive classifications, this can be stated mathematically as:

$$Sensitivity = TP / (TP + FN) \quad [3] \quad (2)$$

- 3) **Specificity:** The proportion in instances that are actually negative from correct negative classifications. This can be mathematically stated as:

$$Specificity = TN / (TN + FP) \quad [3] \quad (3)$$

- 4) **Positive Predictive Value (PPV):** The Proportion in instances that are predicted positive from correct positive classifications. This can be stated mathematically as:

$$PPV = TP / (TP + FP) \quad [3] \quad (4)$$

- 5) **Negative Predictive Values (NPV):** The Proportion in instances that are predicted negative from correct negative classifications. This can be stated mathematically, as:

$$NPV = TN / (TN + FN) \quad [1] \quad (5)$$

Table 5: Comparison of Learning Models Performance

| Model     | DT    | RF  | SVM    |
|-----------|-------|-----|--------|
| Accuracy  | 84.5% | 84% | 91.22% |
| Recall    | 85%   | 84% | 91.15% |
| Precision | 85%   | 84% | 91.00% |
| F1 score  | 85%   | 84% | 91.00% |

Table 5 display the comparison of performance matrix where accuracy, Recall and Precision & F1 score measures used for evaluation of models. Among the learning algorithms used SVM obtained the highest accuracy, recall

and precision and F1-score. SVM obtained 91.22% in accuracy, 91.15% in Recall, 91% for precision and 91% for f1 score. Since the SVM model got the highest accuracy the proponents plotted its learning curve and validation curve.

## 1. Learning Curve graphs

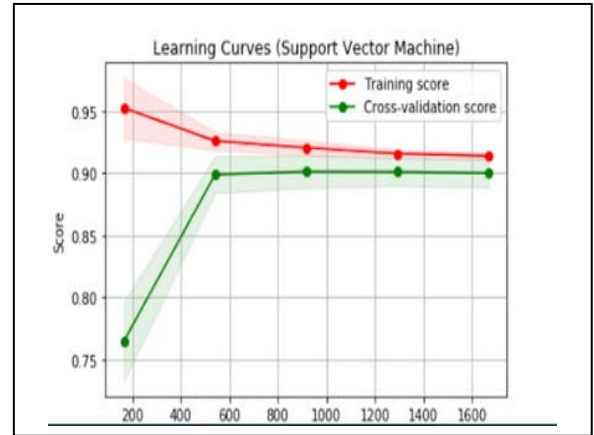


Fig. 2. Learning curve with SVM in gamma 5

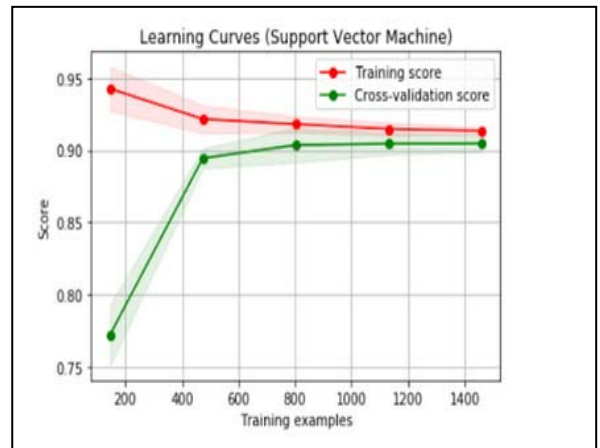


Fig. 3. Learning curve with SVM in gamma4

The figure 2 illustrates the learning curve graph in gamma 5 where the maximum training score mean was 0.952 and the maximum cross validation mean was 0.901, maximum training score was 0.994 and the maximum cross-validation score was 0.811. The learning curve for the training error results was above the one for validation error. The accuracy measure described how good the model is and the MSE on the other side described how bad the model is. The irreducible error gives an upper bound.

## 2. Learning Validation graphs

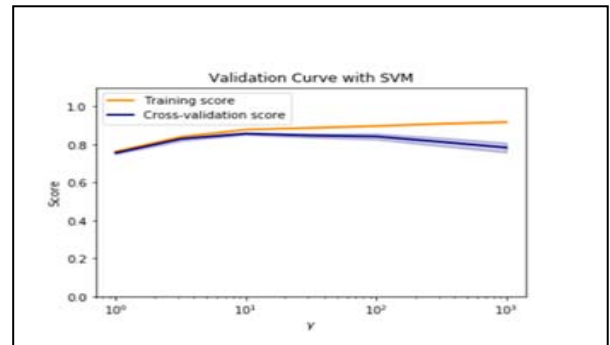


Fig. 4. Validation curve with SVM in gamma 5

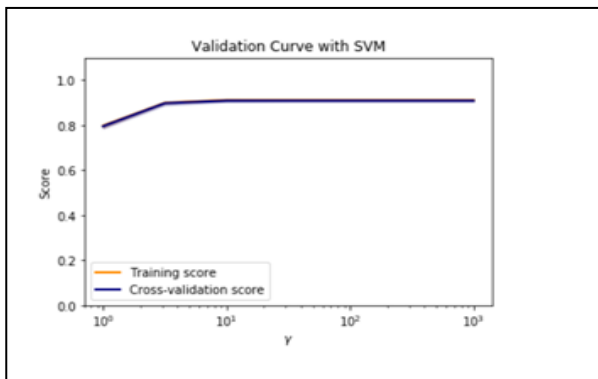


Fig. 5. Validation curve with SVM in gamma 4

Figure 4 shows the validation curve with SVM in gamma 5 where the maximum training R-squared score was 0.918 and the maximum cross-validation score was 0.857 while in figure 5 shows the validation curve with SVM in gamma 4 where the maximum training R-squared score was 0.912 and the maximum cross-validation score was 0.909. It shows that the validation curve with SVM in gamma 5, the gamma is best at 10 to 100.

All models were evaluated, the results of the analysis have been shown in tables 2-5. In training the datasets using different classifiers the proponent had drawn the following figures. When it comes to the accuracy measure, the Support Vector Machine (SVM) obtained 91.22%, it is significantly better than all of the learning algorithms, DT 85%, RF 84%. As shown in Tables 2-5. With respect to Recall measure SVM obtained the Recall of 91.15%, evidently better than all other base classifiers, DT 85%, RF 78%. As shown in Tables 2-5. When it comes to F1score measure, SVM obtained 91% evidently better than all of the base classifiers such as DT 85% and RF 84%.

## V. CONCLUSION AND FUTURE WORK

The Higher Education Institutions (HEIs) becoming more accountable for students career outcomes and as jobs in the labor market increases its competition, institution needs to identify students' employability. This study was a preparation for machine learning approach of predicting students' employability. The preparation stage created a great effect in the results of learning algorithms. Therefore researchers concluded that Support Vector Machine (SVM) produces a predictive model that obtained the highest accuracy of 91.22% and recall measures of .991 or 91.15% and 91% for precision and recall respectively. The researchers realized that gamma is best at 10 to 100 in gamma 5 as shown in figure 3. The researchers concluded that the learning curve and validation curve that it shown was not over fit or under fit. For future work the researchers will analyzed what skillsets will give higher importance score in predicting students' employability. The researcher will use the results of the experiments in real scenario to validate its accuracy. The model will be used in creating a system that will predict students' employability of

engineering students and determine what skillset needs to be improved.

## VI. ACKNOWLEDGMENT

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